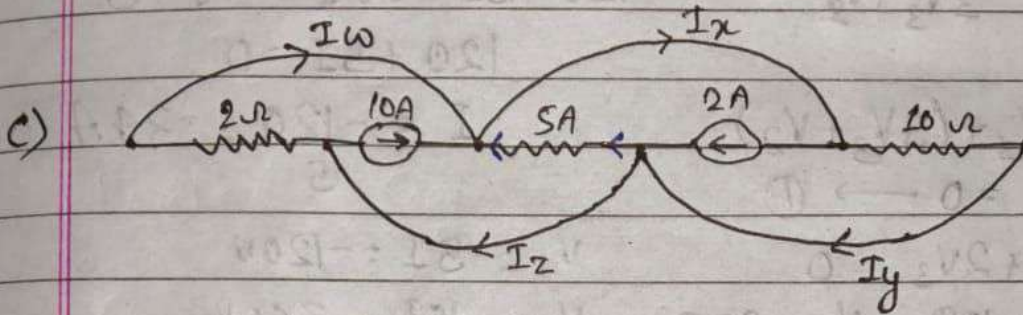


$$I_{2\Omega} = \frac{16}{2} = 8A$$

current entering =
current leaving

$$I + 2A = I_{2\Omega} = 8A$$

$$I = 8 - 2 = 6A$$



$$10 + I_w + I_{5A} = I_x$$

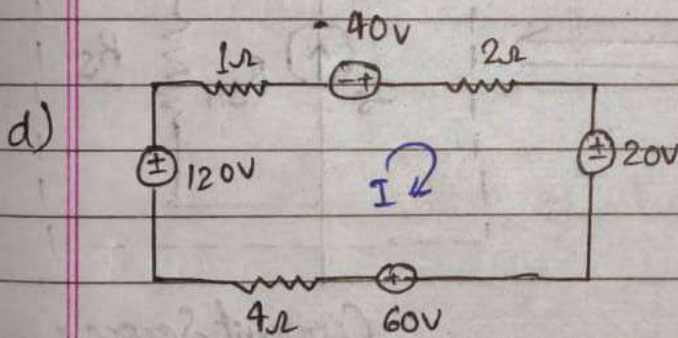
$$I_z = I_w + 10$$

$$\boxed{I_w = 0; I_x = 10A}$$

$$2 + I_y - I_{5A} - I_z = 0$$

$$I_z = I_y + 2$$

$$\boxed{I_y = 10 - 2 = 8A}$$



$$-I(1) + 40 - I(2) - 20 - 60 - I(4) + 120$$

$$(-1-2-4)I + (40+20-60+120) = 0$$

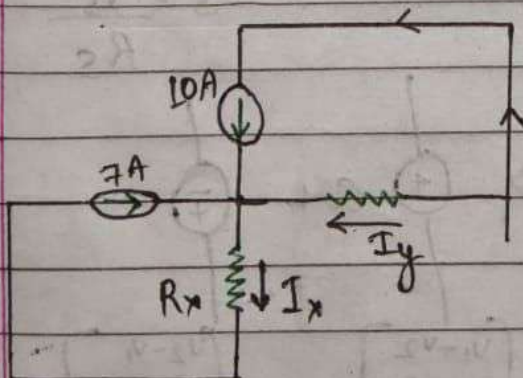
$$-7I + 80 = 0$$

$$7I = 80$$

$$I = \frac{80}{7} \approx 11.43A$$

28/09/26

e)



find the value of I_x and I_y .

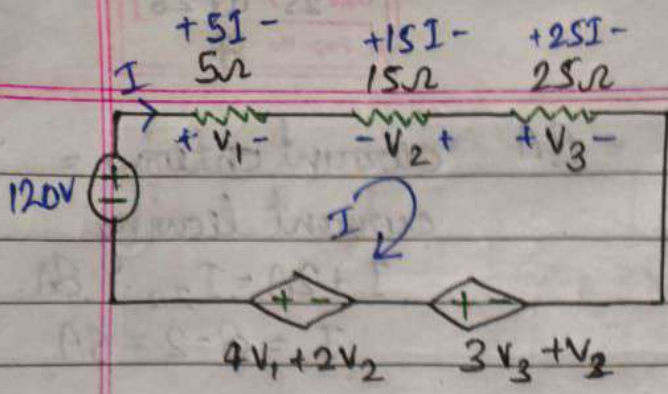
solve →

$$\boxed{I_x = 7A}$$

$$I_y = -10A \Rightarrow 7 + 10 + I_y = I_x$$

$$7 + 10 + I_y = 7$$

$$\boxed{I_y = -10A}$$



find out the value of V_1 , V_2 , V_3 ?

$$120 + 15I - 60I + 50I = 0$$

$$120 + 5I = 0$$

$$I = \frac{-120}{5} = -24A$$

$$V_1 = 5I = -120V$$

$$V_2 = -15I = 360V$$

$$V_3 = 25I = -600V$$

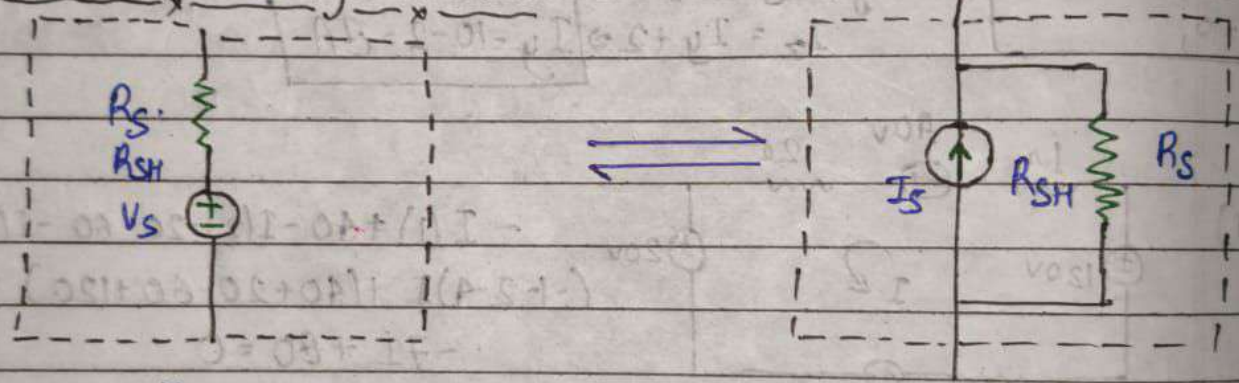
Using
KVL

$$120 - V_1 + V_2 - V_3 + (3V_3 + V_2) + (4V_1 + 2V_2) = 0 \rightarrow (1)$$

$$\Rightarrow 120 + 3V_1 + 4V_2 + 2V_3 = 0$$

$$V_1 = 5I, V_2 = -15I, V_3 = 25I$$

Source Transformation $\rightarrow$

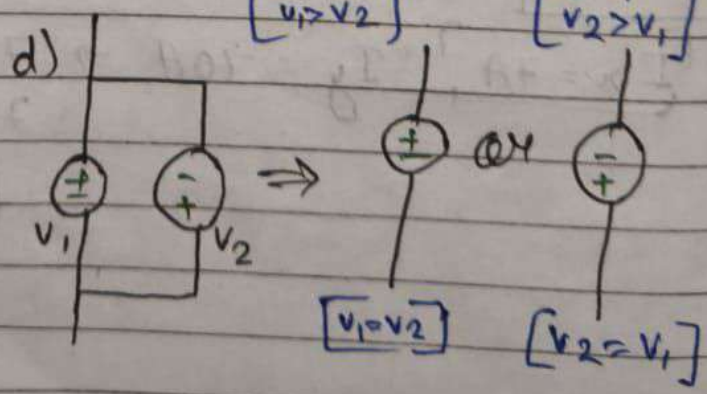
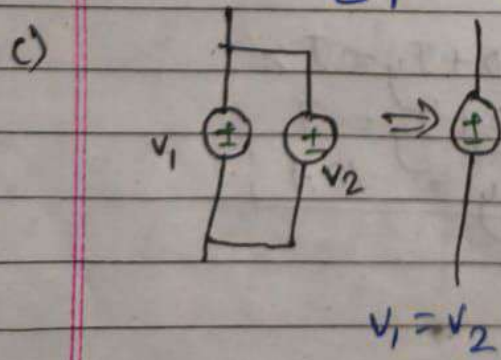
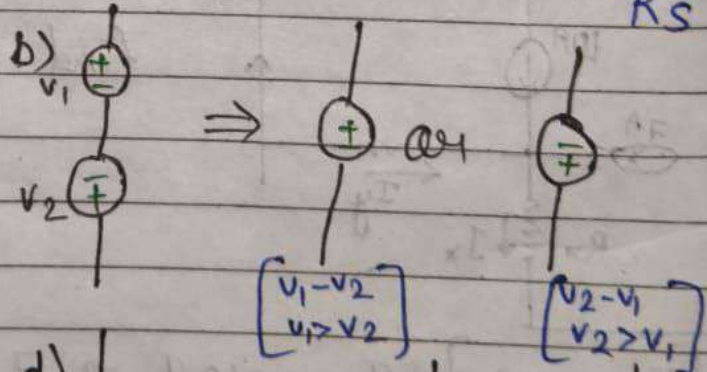
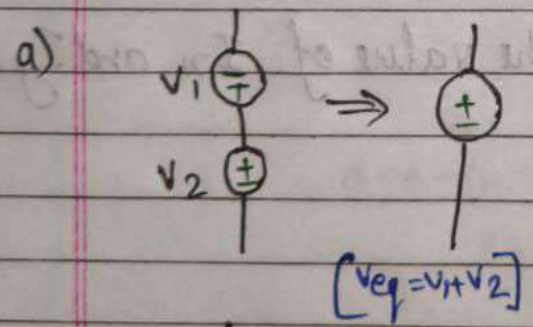


Voltage Source

$$V_S = I_S \cdot R_{SH}$$

Current Source

$$I_S = \frac{V_S}{R_S}$$



a)

c)

Ques

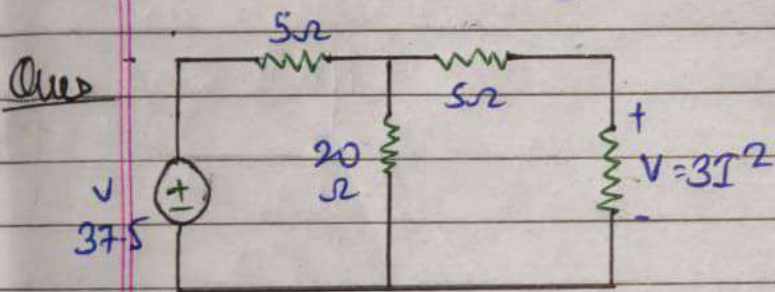
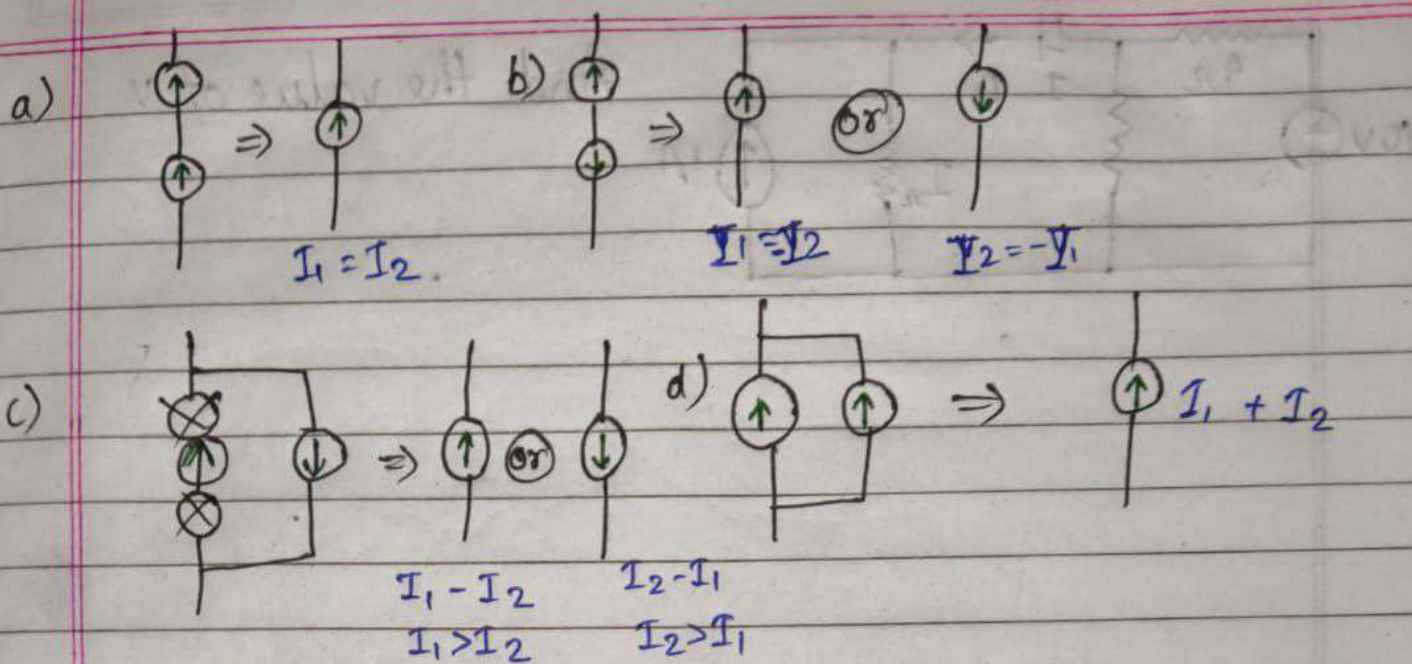
V
37

Ans $\rightarrow$

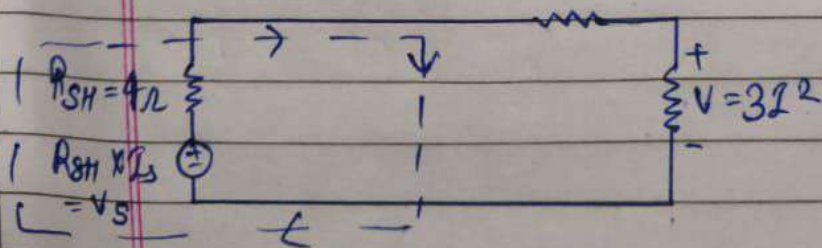
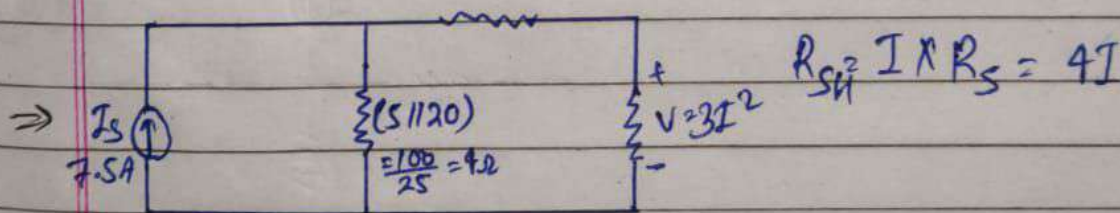
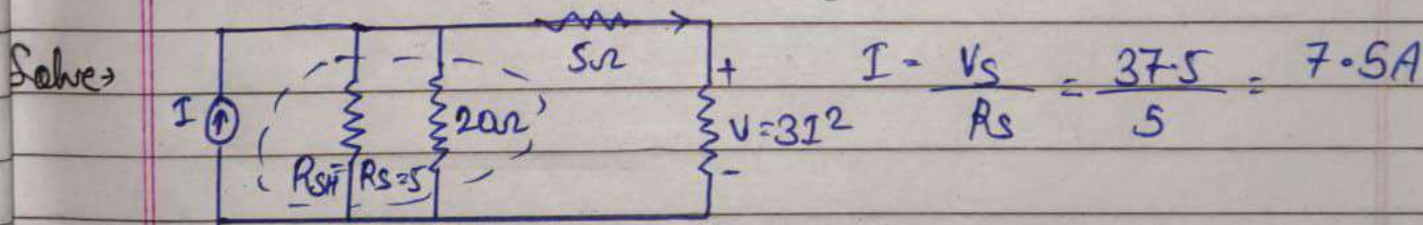
$\Rightarrow$

$R_{SH} =$

$R_{SH} \times$
 $= V_S$



find the value of V using source transformation?



$$30 - 4I - 5I - 3I^2 = 0$$

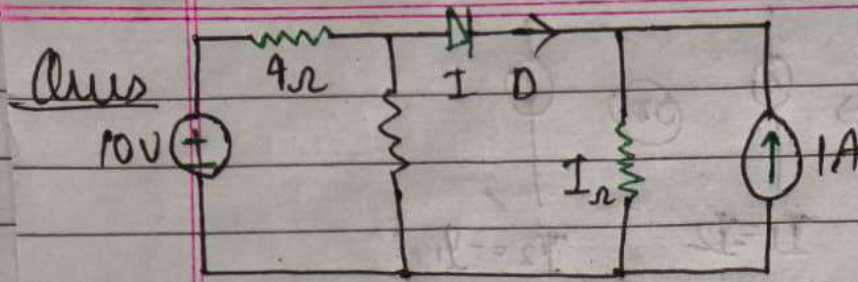
$$3I^2 + 9I - 30 = 0$$

$$I^2 + 3I - 10 = 0$$

$$I^2 + (5-2)I - 10 = 0$$

$$I^2 + 5I - 2I - 10 = 0$$

$$I = -5 \times I = 2, [V = 3I^2 = 12] \leftarrow I(I+5) - 2(I+5) = 0$$



find the value of V ?